HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering

CSE221: Computer Algorithms

Batch: 23rd

Semester: Spring 2024

Date: 09/06/2024 **Mid-Term Examination** Time: 1 Hour 30 minutes

Total Marks: 30

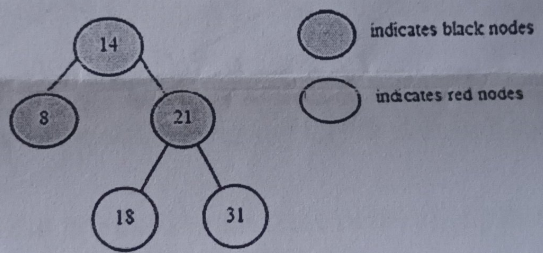
Answer any **five** questions of the followings. $5 \times 6 = 30$

1. (a) Differentiate between Time efficiency and Space efficiency. 2
   (b) Find the complexity from the runtime of Merge Sort given below: 4

$$T(n) = \begin{cases} 1; & if\ n \leq 1 \\ 2 \times T\left(\frac{n}{2}\right) + n\ ; & otherwise \end{cases}$$

2. (a) What are the steps involved in the analysis framework? 2
   (b) **ALGO** is a list of 7 numbers as follows: 4
   59 37 62 78 12 70 57
   Sort the list using Quicksort algorithm.

3. (a) State the **Merge** function in Merge Sort algorithm. 3
   (b) Insert 20, 23 and 19 on the following Red Black tree: 3



4. (a) List out the Steps in Mathematical Analysis of Recursive Algorithms. 2
   (b) Compute the longest common subsequence between A= "USA" and B= "PSBA" using recursion. 4

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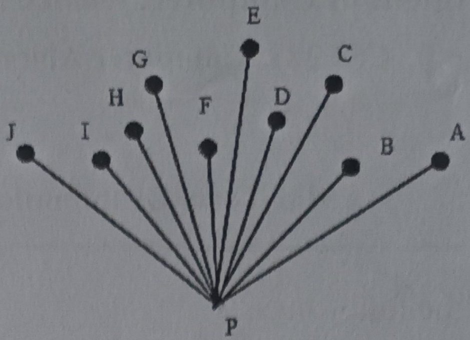
5. (a) Describe the components of Greedy method in an algorithm. 3
   (b) Suppose you have 8 elements in an array as follows: 3

Address	1	2	3	4	5	6	7	8
Elements	5	-8	3	11	15	-17	25	31

Draw the recursion tree to find the maximum and minimum elements from the array using recursive Mini-Max algorithm.

6. (a) Generate the Convex Hull from the given figure using Graham's Scan Algorithm: 3



(b) Suppose, you have four slots and six jobs with the following profits and deadlines respectively: 3

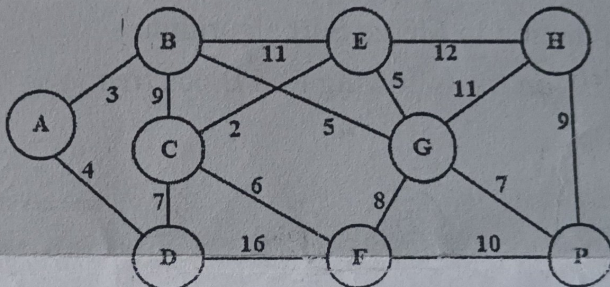
$\{p_1, p_2, p_3, p_4, p_5, p_6\} = \{20, 17, 12, 10, 9, 8\}$ *and*

$\{d_1, d_2, d_3, d_4, d_5, d_6\} = \{3, 1, 4, 1, 3, 2\}$

If each job takes one unit of time to execute, find the optimal solution using Greedy Algorithm.

7. (a) Differentiate between Prim's algorithm and Kruskal's algorithm. 2

(b) Apply Prim's and Kruskal's algorithm to generate the minimum spanning tree for the following graphs: 4



P = [illegible]
K = [illegible]

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 227: Data Communication

Batch: 23rd
Date: 26/06/2024

Mid-Term Examination

Semester: Spring 2024
Time:1 Hour 30 Minutes
Total Marks : 30

Answer any **five** questions: $5 \times 6 = 30$

1. (a) List and explain the five components of a data communications system. 3
   (b) Explain the differences between the duties of the IETF and IRTF. 3

2. (a) What are the three necessary criteria for an effective and efficient network? Explain. 3
   (b) Assume we have created a packet-switched internet. Using the TCP/IP protocol suite, we need to transfer a huge file. What are the advantages and disadvantages of sending large packets? 3

3. (a) Assume a system uses five protocol layers. If the application program creates a message of 100 bytes and each layer (including the fifth and the first) adds a header of 10 bytes to the data unit, what is the efficiency (the ratio of application layer bytes to the number of bytes transmitted) of the system? 2
   (b) Show the encapsulation and decapsulation process in TCP/IP model. 4

4. (a) Explain the Nyquist theorem and Shannon capacity for calculating the data rate. 3
   (b) Explain Multiplexing and De-multiplexing in TCP/IP protocol suite. 3

5. (a) Write the characteristics that are used to measure network performance. 3
   (b) Name the three types of transmission impairment. Explain. 3

6. (a) What are the differences between parallel and serial transmission? 3
   (b) Compare and contrast between Pulse Code Modulation (PCM) and Delta Modulation (DM). 3

7. (a) List and explain five line coding schemes. 3
   (b) Distinguish between signal element and data element. 3

Jitter. Timeliness. Accuracy.

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE 225: Digital Logic Design

Batch: 23rd

Semester: Spring 2024

Date: 24/06/2024 **Mid-Term Examination** Time: 1 Hour 30 minutes

Total Marks: 30

Answer any **five** questions of the following. $5 \times 6 = 30$

1.	(a)	Explain the difference between a sequential circuit and a combinational circuit.	CLO1	3
	(b)	Apply Boolean identities and simplify this function F (A, B, C) = (A + B' + C') (A + B' + C) (A + B + C').	CLO3	3
2.	(a)	What is the working principle of full adder?	CLO5	2
	(b)	Design a 3-bit binary Adder-Subtractor circuit with proper explanation.	CLO5	4
3.	(a)	Implement the following Boolean equation using only NAND gates Y=AB+CDE+F	CLO5	3
	(b)	Implement three Basic Gates using any Universal Gate.	CLO4	3
4.	(a)	What is a digital computer? Explain the working principle of digital computers with a block diagram.	CLO1	3
	(b)	What will be the SOP (Sum of Product) and POS (Product of Sum) form for this following function F = A + BC' + A(B+AC).	CLO4	3
5.	(a)	What is Latch? Explain with proper diagram.	CLO5	2
	(b)	What is a flipflop? Why is it used over Latch? Make a comparison between latch and flipflop.	CLO5	4
6.	(a)	What is the trigger in Sequential Circuit? Briefly explain its type and use cases.	CLO1	3
	(b)	Construct a Full Adder using the concept of half adder.	CLO4	3
7.	(a)	Write the advantages of Tabulation method over K-Map method.	CLO3	2
	(b)	$F(A, B, C, D) = \Sigma m(0, 1, 3, 7, 8, 9, 11, 15) + \Sigma d(5, 6, 13, 14)$ Minimize the given Boolean function using K-map.	CLO3	4

HAMDARD UNIVERSITY BANGLADESH

In Pursuit of Knowledge

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE223: Object Oriented Programming in Java

Batch: 23rd

Semester: Spring 2024

Date: 11/06/2024 **Mid Term Examination** Time: 1 hour 30 minutes

Total Marks: 30

Answer any **five** questions from the following: 5x6= 30

1	(a)	List some familiar instances where Java is applied in practical contexts.	2
	(b)	Design a Java program that takes a year as input and uses the ternary operator to determine whether the year is a leap year or not. Display an appropriate message accordingly.	4
2	(a)	Summarize the key principles that make Java a secured language.	2
	(b)	Create a Java class named Computer with instance variables brand, processor, and ram. Implement two constructors: a parameterized constructor that initializes the instance variables and another constructor that takes only the brand parameter and invokes the parameterized constructor using the 'this' keyword to set default values for processor and ram.	4
3	(a)	Provide an overview of how the 'this' keyword is employed in Java programming.	2
	(b)	Write a Java program to check whether the number is Armstrong or not.	4
4	(a)	Differentiate between primitive and non-primitive data types.	2
	(b)	Find the output of following two code segments:	2×2 = 4

11 + 12 + 11 - 10 · 10

i.
```
public class Test {
public static void main(String[] args) {int x = 10;
int y = x++ + ++x + --x - x-- + x;System.out.println(y);
}}
```

ii.
```
public class Test {
public static void main(String[] args) {String s1 = "Java";
String s2 = "Java";
String s3 = new String("Java");if (s1 == s2 && s1 == s3) {
System.out.println("Same");}
else
{
System.out.println("Different");}
}
}
```

5	(a)	Compare the functionalities of JDK, JRE, and JVM in the Java ecosystem.	3
	(b)	Create a Java program that prompts the user to enter a direction	3

abbreviation (N, S, E, W). Use a switch statement to convert the abbreviation to its corresponding full name (North, South, East, West) and display it.

6 (a) Explain the rationale behind the do-while loop being referred to as an exit-controlled loop. 2

(b) Create a Java program that factors a given number using a loop. Prompt the user to enter a positive integer, and then compute and display its prime factors. 4

7 (a) Outline the access specifiers and their functions in Java. 2

(b) Create an abstract class called Employee with abstract methods calculateSalary() and displayDetails(). Implement concrete subclasses like HourlyEmployee and SalariedEmployee that extend Employee and provide their own implementations for the calculateSalary() and displayDetails() methods. 4

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
ENS221: Environmental Science
Mid-term Examination

Date: 29/06/2024

Semester: Spring 2024
Time: 1 Hour 30 Minutes
Total Marks: 30

Part A (Answer any Two)

			Marks	CLO
1.	a)	'Environmental Science covers four major areas'. Explain.	3	CLO1
	b)	Mention the importance of Environmental Science in the context of Bangladesh as well as global.	4	CLO1
	c)	'The classroom of Hamdard University Bangladesh is as like as greenhouse. Justify it.	3	CLO1
2.	a)	The Ozone Layer acts as a protective layer. Justify it.	2.5	CLO1
	b)	Explain how the burning of fossil fuels contributes to climate change.	4.5	CLO1
	c)	Give a description of eutrophication in your own words.	3	CLO1
3.	a)	The photosynthesis process: $CO_2 + H_2O \rightarrow C_6H_{12}O_6 + O_2$. It is helps to maintain a relationship between plant and animal. Elucidate it.	3.5	CLO1
	b)	Global warming is a part of climate change', Justify it.	3	CLO1
	c)	Why most weather phenomena, like clouds and storms, occur in the troposphere.	3.5	CLO1

Part B (Answer any One)

			Marks	CLO
4.	a)	What do you think about the main idea of 'Sustainable Development'? Write the historical background (1957-2015) of it.	5	CLO2
	b)	How do heat waves affect people? What should you do if someone is experiencing a heat-related illness?	5	CLO2
5.	a)	'Bangladesh is a flood prone country'. Explain	2.5	CLO2
	b)	Although cyclones have always been a threat to Bangladesh, some research indicates that their frequency has increased recently. Examine the reasons behind the daily rise in the number of cyclones. Give your reasoning to back up this assertion.	2.5	CLO2
	c)	Bangladesh's vast low-lying coastal zones make it very vulnerable to increasing sea levels. Examine the possible effects of rising sea levels on Bangladesh's coastal areas.	5	CLO2